

MATH-3108-103-70510-2022SP-Linear Algebra I Exam 2

Logan Mills

TOTAL POINTS

72 / 100

QUESTION 1

1 Question 1 0 / 3

✓ - 3 pts Incorrect. The correct answer is $\det A = 0$.

QUESTION 2

2 Question 2 0 / 6

✓ - 6 pts Both incorrect. The correct answers are C and D.

QUESTION 3

3 Question 3 3 / 3

✓ - 0 pts Correct

QUESTION 4

4 Question 4 3 / 3

✓ - 0 pts Correct

QUESTION 5

5 Question 5 0 / 3

✓ - 3 pts Incorrect. The correct answer is False.

QUESTION 6

6 Question 6 0 / 3

✓ - 3 pts Incorrect. The correct answer is iv.

QUESTION 7

Question 7 10 pts

7.1 Question 7a 5 / 5

✓ - 0 pts Correct

7.2 Question 7b 5 / 5

✓ - 0 pts Correct

QUESTION 8

8 Question 8 7 / 7

✓ - 0 pts Correct

QUESTION 9

9 Question 9 12 / 12

✓ - 0 pts Correct

QUESTION 10

10 Question 10 2 / 10

✓ - 2 pts Did not show that the zero vector is in the set

✓ - 3 pts Did not show closure of + -

✓ - 3 pts Did not show * closure

QUESTION 11

11 Question 11 4 / 8

✓ - 4 pts Do not show understanding of $(AB)^{-1} = B^{-1} A^{-1}$

QUESTION 12

12 Question 12 9 / 10

✓ - 1 pts Not simplified

QUESTION 13

13 Question 13 6 / 6

✓ - 0 pts Correct

QUESTION 14

Question 14 16 pts

14.1 Question 14a 6 / 6

✓ - 0 pts Correct

14.2 Question 14b 10 / 10

✓ - 0 pts Correct

Name: Logan Mills

Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
March 16, 2022

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, **NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!**
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

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Multiple Choice and True/False. Fill in the box corresponding to the correct answer(s).

1. (3 points) Let A be a 4×4 matrix. If A has exactly 3 pivot positions, then which of the following statements is true? (Hint: Only one of the following statements is true.)
- ☐ A is an invertible matrix ☒ The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
- ☐ $\det A = 0$ ☐ The null space of A contains only the zero vector
2. (6 points) Let A and B be $n \times n$ matrices. Select the two *false* statements.
- ☒ a) $(AB)^T = B^T A^T$ ☒ b) $(A+B)^T = A^T + B^T$
- ☐ c) $(A^T)^{-1} = A$ ☐ d) $(AB)^{-1} = A^{-1}B^{-1}$
- ☐ e) $\det A^T = \det A$ ☐ f) $\det(AB) = (\det A)(\det B)$
3. (3 points) A basis for a vector space V is a _____ of vectors in V .
- ☐ a) minimal spanning set ☐ b) maximal linearly independent set
- ☒ c) both (a) and (b) ☐ d) neither (a) nor (b)
4. (3 points) The zero vector in a vector space is unique.
- ☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
5. (3 points) The determinant of a triangular matrix is the sum of the entries on the main diagonal.
- ☒ a) True ☐ b) False
6. (3 points) Consider the matrix $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$. The vector $\begin{bmatrix} 0 \\ 1 \\ -2 \\ 0 \end{bmatrix}$ is in _____.
- ☐ i) $\text{Col } A$ ☒ ii) $\text{Nul } A$
- ☐ iii) both $\text{Col } A$ and $\text{Nul } A$ ☐ iv) neither $\text{Col } A$ nor $\text{Nul } A$

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Open Answer. Show ALL your work!

7. (5 points each) Consider the matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

a) Calculate A^{-1} . $4 - 6 = -2$

$$\frac{1}{-2} \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 3/2 \\ 1 & -1/2 \end{bmatrix}$$

b) Use the result of part (a) to solve the system $A\mathbf{x} = \mathbf{b}$ where $\mathbf{b} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$.

Solutions using other methods will receive no credit.

$$\mathbf{x} = A^{-1} \mathbf{b}$$

$$\begin{bmatrix} -2 & 3/2 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \end{bmatrix}$$

8. (7 points) Is the set $W = \left\{ \begin{bmatrix} -a \\ a - 6b \\ 2b + a \end{bmatrix} : a, b \text{ real} \right\}$ closed with respect to scalar

multiplication?

Hint: There is a very quick "clever" way to do this problem, but the standard method also works.

$$\begin{bmatrix} -1 & 0 \\ 1 & -6 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

The set is closed with respect to scalar multiplication because all matrix entries have variables a or b , so a or b can be any scalar number.

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9. (12 points) Find the LU -factorization of the matrix $A = \begin{bmatrix} -5 & 0 & 2 \\ 15 & 3 & -3 \\ 15 & 18 & 20 \end{bmatrix}$.

$$\begin{array}{l} 3R_1 + R_2 \rightarrow R_2 \\ 3R_1 + R_3 \rightarrow R_3 \end{array} \begin{bmatrix} -5 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 18 & 26 \end{bmatrix} \begin{array}{l} -6R_2 + R_3 \rightarrow R_3 \end{array}$$

$$U = \begin{bmatrix} -5 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 8 \end{bmatrix}$$
$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -3 & 6 & 1 \end{bmatrix}$$

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10. (10 points) Show that the set of all polynomials of the form $p(t) = a + bt^2$ where a and b are real is a subspace of $\mathbb{P}_2$.

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ t^2 \end{bmatrix} \right\} \quad n=2 \rightarrow \text{sub of } \mathbb{R}^n$$

$p(t) = a + bt^2$ is a subspace of $\mathbb{P}_2$ because $p(t)$ is a second degree polynomial, so if a and b are real numbers the set is closed and $p(t)$ is a subspace.

11. (8 points) Given $A^{-1} = \begin{bmatrix} -4 & -2 & 3 \\ -2 & 1 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} -4 & -9 & 5 \\ -1 & -4 & 2 \\ 6 & 15 & -8 \end{bmatrix}$ for invertible matrices A and B , calculate $(AB)^{-1}$.

$$c_{11} = 16 + 2 + 18 = 36$$

$$c_{21} = 8 + 0 - 1 = 7$$

$$c_{31} = -12 + 0 - 6 = -18$$

$$c_{12} = 36 + 8 + 45 = 89$$

$$c_{22} = 18 - 4 + 0 = 14$$

$$c_{32} = -27 - 15 = -42$$

$$c_{13} = -20 - 4 - 24 = -48$$

$$c_{23} = -10 + 2 = -8$$

$$c_{33} = 15 + 8 = 23$$

$$(AB)^{-1} = \begin{bmatrix} 36 & 89 & -48 \\ 7 & 14 & -8 \\ -18 & -42 & 23 \end{bmatrix}$$

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12. (10 points) Consider the system of equations
$$\begin{cases} 2x - 3y + 4z = 1 \\ x + 6z = 0 \\ 3x - 2y = 5 \end{cases}$$

Given that $\begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ 3 & -2 & 0 \end{vmatrix} = -38$, use Cramer's Rule to solve for the variable x only.

Place your answer in the box provided.

$$\begin{vmatrix} 1 & -3 & 4 \\ 0 & 0 & 6 \\ 5 & -2 & 0 \end{vmatrix}$$

-38

$$1(0 + 12) + 0 + 5(-18 - 0)$$

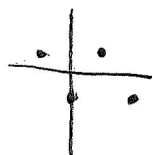
$$12 - 90 = -78$$

-38

$$= \frac{78}{38}$$

$$x = \boxed{\frac{78}{38}}$$

13. (6 points) Find the area of the parallelogram whose vertices are $(0, -2)$, $(5, -2)$, $(-3, 1)$, and $(2, 1)$. Place your answer in the box provided.



$$\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix}$$

$$\text{Area} = \boxed{15}$$

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14. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (6 points) Find a basis for the column space of A .

$$\text{Col } A = \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ -13 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ -12 \\ -10 \end{bmatrix} \right\}$$

- b) (10 points) Find a basis for the null space of A .

$$x_1 = 2x_2 - 3x_5 - x_6$$

$$x_3 = -x_5$$

$$x_4 = 0$$

$$x_2, x_5, x_6 \text{ free}$$

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} x_5 + \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} x_6$$

$$\text{Nul } A = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$